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Stuttgart | Shanghai



Summary

- Builder with hands-on experience in robotics, applied ML, system integration, and human-robot interaction.

Technical Skills

- Software & Data: Python, C#, MongoDB, Git
- AI & ML: PyTorch, Deep Learning, Temporal Modeling (LSTM), RAG, YOLO
- Integration: UDP, ROSBridge, Unity/MRTK
- Analysis & Prototyping: Power BI, DAX, Grasshopper, Rhino, Figma

Professional Experience

STRABAG — 2025.03 - 2026.10
Working Student (Data Services) — Stuttgart, Germany

Product Analytics | Product Ownership

- Developed and maintained an internal analytics dashboard for monitoring user adoption, content usage, and engagement.
- Built a Python ETL workflow connecting MongoDB to Power BI, automating daily processing and refresh of 10k+ records.
- Defined product metrics and translated stakeholder needs into data requirements.

RoboticPlus.AI (Shanghai) Co., Ltd — 2022.08 - 2024.09
Robotic Solution Engineer — Shanghai, China

Robotics Application | Solution Engineering

- Translated application requirements into robotic systems integrating industrial robots, sensing, custom hardware, and software components.
- Developed and integrated prototypes using Python, C#, Grasshopper, Unity/MRTK, depth sensing, and robot communication interfaces.
- Iteratively refined robotic workflows through rapid prototyping and testing, coordinating hardware, software, and on-site deployment constraints to deliver working systems.

Key Projects

Master's Thesis Project
Proactive Robotic Assistance for Interior Construction Assembly

Individual Focus: Human-Robot-Collaboration | System Integration | Deep Learning

- Built a human–robot collaboration system integrating **human activity recognition**, **decision making**, **speech interaction**, and **robot execution** to enable proactive assistance based on human task step and progress.
- Trained and iteratively optimized skeleton-based LSTM models for task step classification and progress prediction, addressing class imbalance and generalization.
- Designed an event-driven orchestration layer combining model outputs, human responses, task context, and robot states to coordinate robot actions through Python, UDP, and ROSBridge.
- Integrated and tested the UR10e ceiling-assembly demonstrator and contributed to the user study.

GitHub: https://github.com/Lo122/hrc_communication | https://github.com/Lo122/LSTM_HRC

Seminar Project

GenGarden: Human–AI–Robot Sand Painting Co-Creation

Individual Focus: Generative AI | Multi-Agent Workflow

- Built a human–AI–robot sand-painting workflow integrating AI image generation, UR robot execution, and iterative visual evaluation.
- Developed the system MVP and integrated camera sensing, image-generation agents, Grasshopper, robot control, and evaluation modules.
- Structured Gemini outputs into robot-executable data and implemented a **generate - execute - evaluate - rework** loop that automatically triggers reprocessing based on evaluation results.

GitHub: https://github.com/Lo122/zen_garden

Plan Navigator | RAG-based Planning Assistant

AI Product Prototyping | LangChain | AI-Assisted Development

- Built a RAG-based planning assistant that turns project updates, blockers, and feedback into structured weekly action plans.
- Implemented document ingestion, semantic retrieval, context construction, and LLM generation using Python and Chroma, with FastAPI exposing the core workflow.

GitHub: https://github.com/Lo122/Plan_Navigator

Education

University of Stuttgart (M.Sc.) — 2024.10 - 2026.10
- Integrative Technologies and Architectural Design Research (ITECH) — Stuttgart, Germany

Zhejiang University (B.Arch.) — 2017.09 - 2022.06
- Architecture — Hangzhou, China

Other Skills

- Languages: Mandarin (native) | English (fluent)

Publication

CAADRIA 2022:

Guo, Y., **Luo, Y.,** Wang, S., Tan, Y.Y., & Tracy, K. Robotic Fabrication of Topology Optimised Concrete Components with Reusable Formwork.